Robotic Exoskeleton for Load Transportation

IMU-based locomotion intent recognition and assist gating in MATLAB

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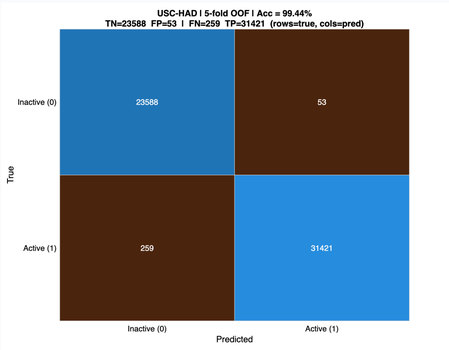
Key message: the validated baseline is a reproducible HuGaDB (GitHub v1 corpus) MATLAB assist-gating pipeline that now respects official corruption metadata and streaming session structure: canonical six-IMU windows from realistic `various` multi-activity sequences, alongside controlled single-activity variants such as walking/running/sitting/standing, feed 30-D features, a binary active-vs-inactive SVM, and FSM hysteresis that turns noisy window predictions into a stable assist on/off command.

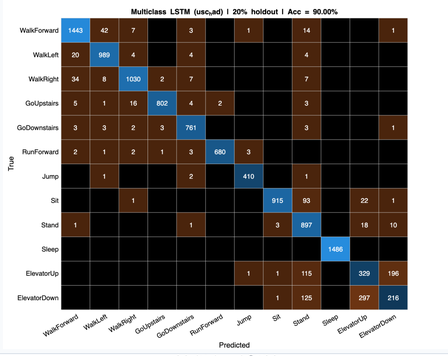
Binary SVM, 5-fold OOF	Binary SVM, 5-fold OOF	Multiclass SVM, 5-fold OOF	Multiclass SVM, 5-fold OOF	Multiclass LSTM, 20% holdout	Multiclass LSTM, 20% holdout
95.89%	**99.44%**	**80.80%**	**53.41%**	**82.03%**	**90.00%**
HuGaDB	USC-HAD	HuGaDB	USC-HAD	HuGaDB	USC-HAD

USC-HAD

Confusion matrices and LSTM training (`trainNetwork` logs)

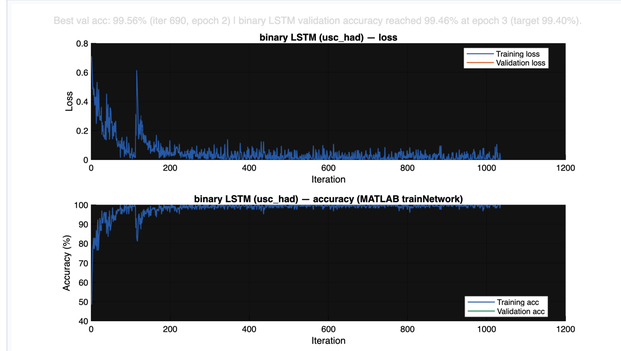


Binary SVM

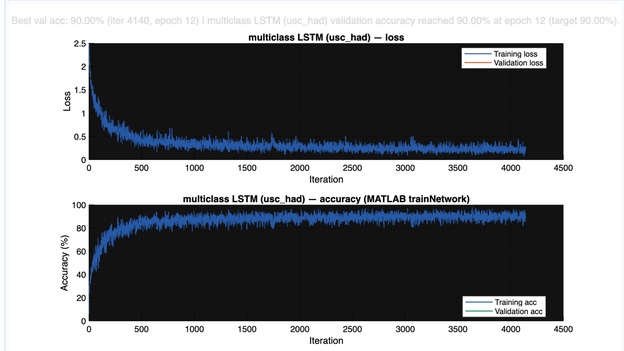


Multiclass LSTM

Per-iteration logs (same quantities as the console): `results/logs/binary/lstm_binary_training_usc_had.txt` and `results/logs/multiclass/lstm_multiclass_training_usc_had.txt`.



Binary LSTM — loss / accuracy



Multiclass LSTM — loss / accuracy

How to read the percentages

Every percentage on this poster is a **classification accuracy**: the share of IMU windows whose predicted label matches the dataset ground truth. For example, **95.89%** means about **96 out of every 100 windows** were labeled correctly.

- **5-fold OOF** means five-fold out-of-fold cross-validation: the model is repeatedly trained on 80% of the data and tested on the remaining 20%, so every window is evaluated once while unseen during its own test fold.
- **20% holdout** means one fixed split: 80% of the windows are used for training and 20% are kept aside for a final test.
- **Binary** percentages measure correct *active vs. inactive* decisions; **multiclass** percentages measure correct assignment to the dataset's native activity class.

Problem and approach

- Manual load handling above 20 kg raises fatigue and musculoskeletal injury risk.
- An exoskeleton is useful when assistance engages and releases at the right time.
- This work targets a **software supervisor**: infer active movement from IMUs and issue a reliable assist on/off signal without EMG.
- The goal is not full torque control yet; it is a robust decision layer that can sit above future hardware control.

Implemented workflow

- **Input:** canonical GitHub v1 raw under `v1_cleanup_github/` for training and held-out replay for simulation; a second public benchmark provides comparison under the same feature layout.
- **Dataset handling:** loader applies the official gyro-corruption rules at read time, distinguishes `single_activity` from `multi_activity_sequence` sessions, builds `hugadb_dataset.mat`, and records quality metadata so invalid sessions/windows are skipped rather than silently used.
- **Session variation coverage:** controlled single-activity recordings include `walking`, `running`, `sitting`, `standing`, `bicycling`, `sitting_in_car`, and `down_by_elevator`; mixed `various` recordings are treated as the streaming-style multi-activity sequences.
- **Fusion:** Kalman filtering is optional and used for a replay-time segment-pitch visualization trace, not for classifier preprocessing.
- **Windowing:** 1.0 s windows at 100 Hz with 50% overlap.
- **Features: 30-D** handcrafted vector, five statistics per IMU slot across six slots.
- **Primary model:** RBF SVM for **binary gating**: active-movement windows vs. inactive windows, using dataset-specific activity-to-binary maps.
- **Supervisory smoothing:** FSM hysteresis, 3 active predictions to engage assist and 5 inactive predictions to release.

Main validated result

The strongest binary-SVM claim in the current repository is the **streaming-policy** model on multi-activity sessions: **95.89%** five-fold out-of-fold accuracy.

Training data	OOF acc.
USC-HAD	**99.44%**
HuGaDB streaming	**95.89%**

This table is kept as **benchmark context**. The final project path does **not** merge datasets; training uses the primary corpus above and keeps replay-based simulation as a separate step.

Data and evaluation protocol

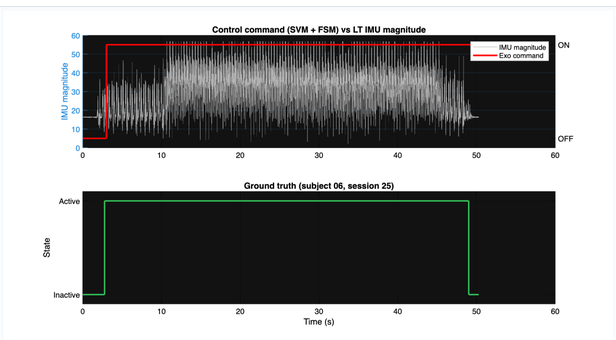
- The final pipeline uses six on-body IMU streams per window, full native class coverage, and **multi-activity sequence sessions** as the realistic streaming policy; single-activity sessions are controlled subsets (walking, running, sitting, standing, bicycling, sitting-in-car, elevator descent).
- The canonical raw layout is `v1_cleanup_github/` (flat `HuGaDB_v1_*.txt`); the loader threads optional manifest metadata into the cache for validation and rebuild traceability.
- The second benchmark uses one IMU stream under the same 30-D layout for cross-dataset comparison.
- The **binary SVM** is the main baseline; the active checkpoint is trained under the held-out-subject policy used by replay evaluation.
- **Binary** asks a simple control question: should assist stay **off** or turn **on**? **Multiclass** asks a harder recognition question: which exact activity is occurring?
- **Multiclass SVM** (12 classes per benchmark) and **binary/multiclass LSTM** are supporting baselines; the SVM uses stratified subsamples and the LSTMs use separate stratified 20% holdouts, so neither is a direct deployment comparison.

Pipeline at a glance

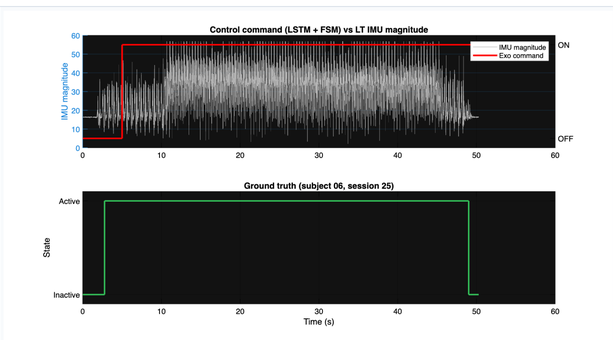
1. Train pool and held-out replay → 2. Optional Kalman visualization trace → 3. 30-D features per window → 4. Binary SVM prediction → 5. FSM hysteresis for assist → 6. Stable on/off command

Replay outputs (HuGaDB subject 06, session 25)

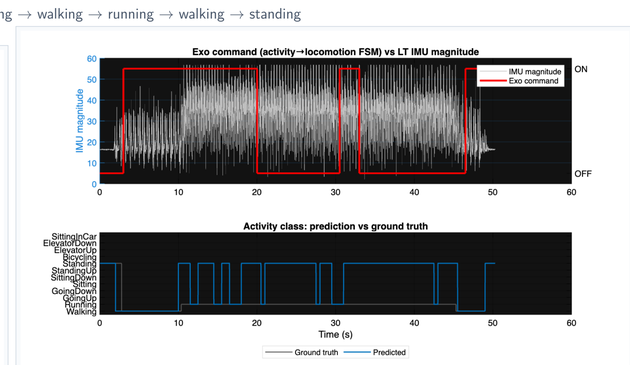
Native activity sequence (file header): standing → walking → running → walking → standing



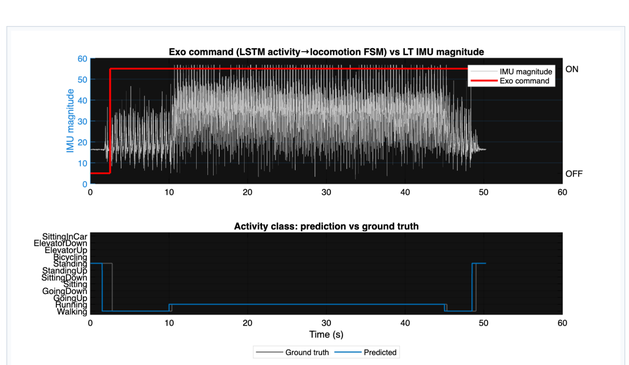
Binary SVM replay: Kalman + FSM assist



Binary LSTM replay (held-out pipeline)



Multiclass SVM replay



Multiclass LSTM replay

Take-home message: the repository now delivers a traceable MATLAB baseline for IMU-based binary assist gating, centered on canonical GitHub v1 training data, benchmark comparison tables, replay-based simulation, and FSM-stabilized control output.

How to read the supporting baselines

- The pipeline plot shows why the FSM matters: it turns noisy window-level predictions into a smoother and more usable assist signal.
- The **binary** task is easier because several activities are merged into two groups; the **multiclass** task is harder because it must separate the exact activity label.
- Multiclass SVM OOF scores (**53.41%** / **80.80%**) are informative for analysis, but not yet strong enough for fine-grained control.
- Multiclass LSTM holdout (**90.00%** / **82.03%**) is a summary metric across distinct native label spaces.

Current limits

- No hardware-in-the-loop or human-subject load trials are reported here.
- End-to-end latency still needs direct benchmarking during future hardware-integrated validation.
- Current output is **assist gating**, not full torque scheduling or terrain-aware control.
- Multiclass and LSTM baselines still need a stronger evaluation protocol aligned with the main binary SVM path.

Next steps

- Align multiclass and LSTM evaluation with the same level of rigor used for the main binary SVM baseline.
- Add hardware integration, latency measurement, and load-carrying trials.
- Move from binary assist gating toward subject-specific, activity-aware assistance.

Summary

- **Delivered:** canonical GitHub v1 data flow, protocol-aware binary SVM, optional Kalman visualization, FSM smoothing, reproducible metrics, and replay in MATLAB.
- **Best validated result:** binary SVM **95.89%** five-fold OOF on the streaming split.
- **System role:** simulation-level assist gating supervisor.
- **Value:** a credible, traceable software baseline that separates benchmark comparisons from the final deployed method.

GitHub